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Team 64: TI Solar Battery Pack Bi-Weekly Update 4

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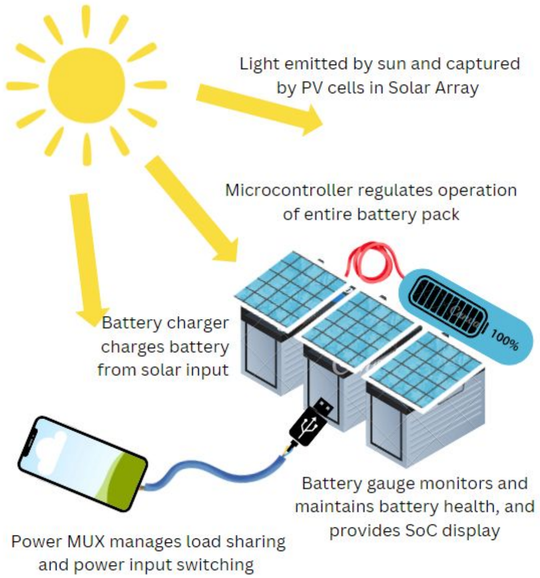
TA: Ali Alenezi

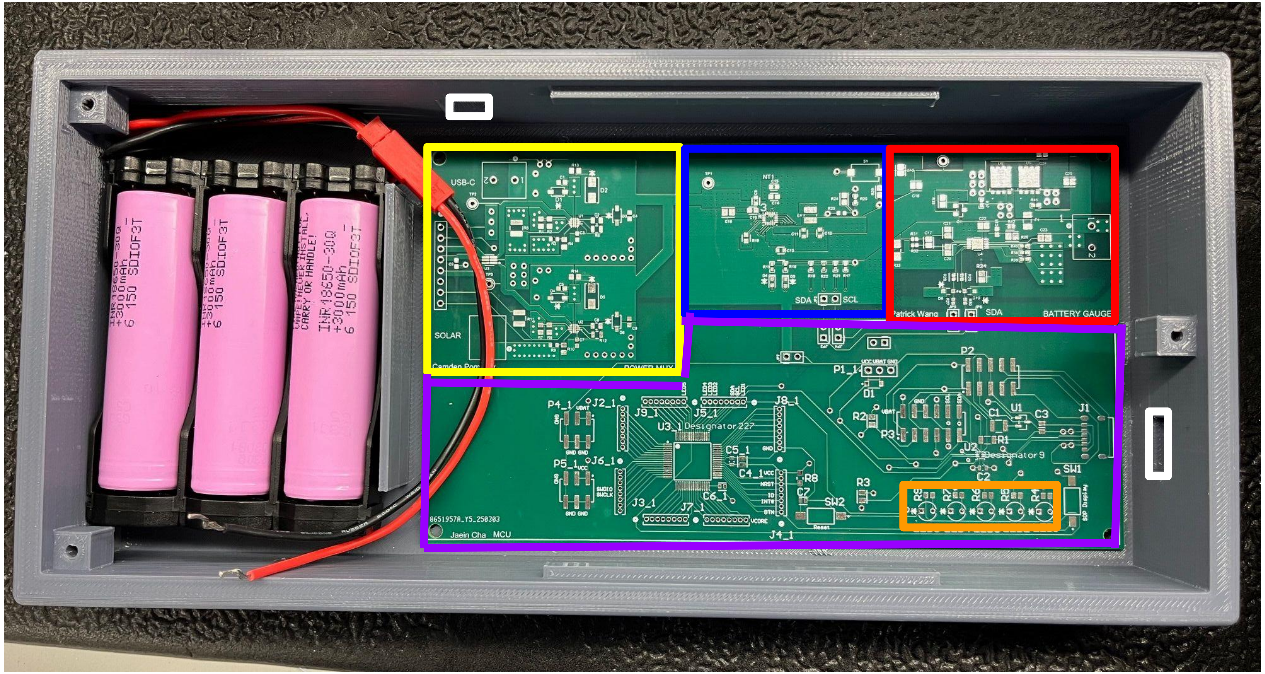
Project Summary

- Natural disasters can cause unexpected power outages resulting in the inability to charge cell phones and other necessary small electronic devices.
- A solution is to create a solar powered battery pack that charges itself from solar panels, and distributes the power to electronic devices through a USB connection.

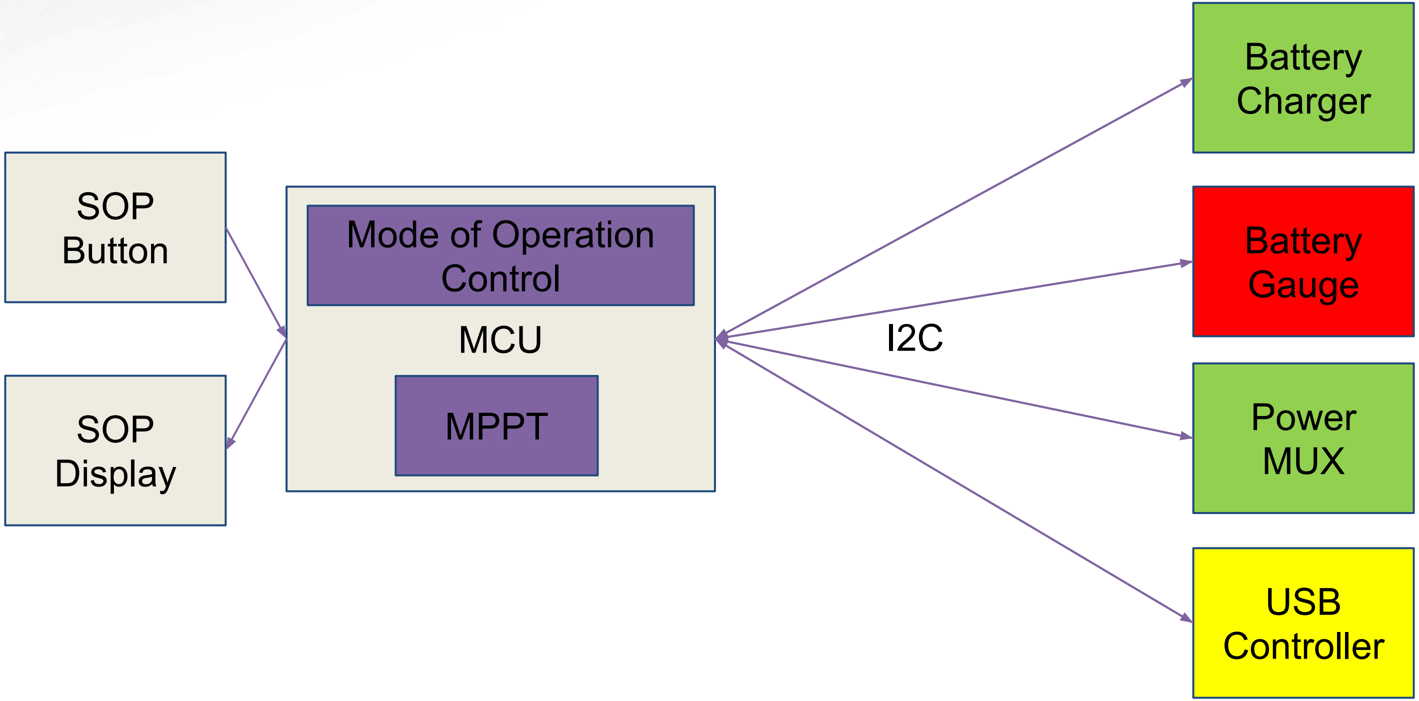


Integrated System Diagram

Yellow - Power Mux
Blue - Battery Charger
Red - Battery Gauge
Purple - MCU
Orange - SOC Display
White - Power Ports



Integrated System Diagram (software)



Project Timeline

Subsystem Designs and Testing (completed 2/3)	Integration of High Power Path (completed 2/17)	Final Integration (completed 3/12)	System Test (to complete by 3/24)	Validation (to complete by 3/31)	Demo and Report (to complete by 4/7)

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Patrick Wang - Battery Gauge

Accomplishments since last update 26 hours	Ongoing progress/problems and plans until the next presentation
• Fully integrated board and parts arrived • Completed integration with the battery charger on new integrated board • Calibrated battery gauge charge and discharge “learning cycle” on old subsystem board	• Fix problem with battery gauge I2C connection • Calibrate new battery gauge, voltage thresholds, temperature

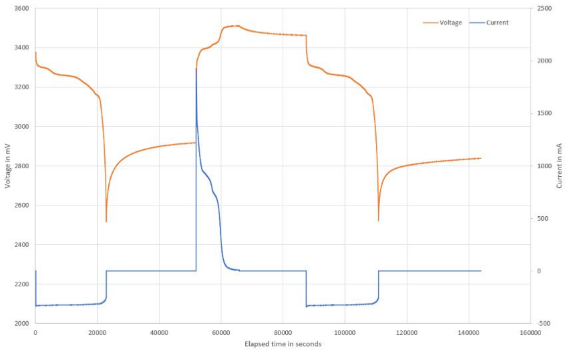
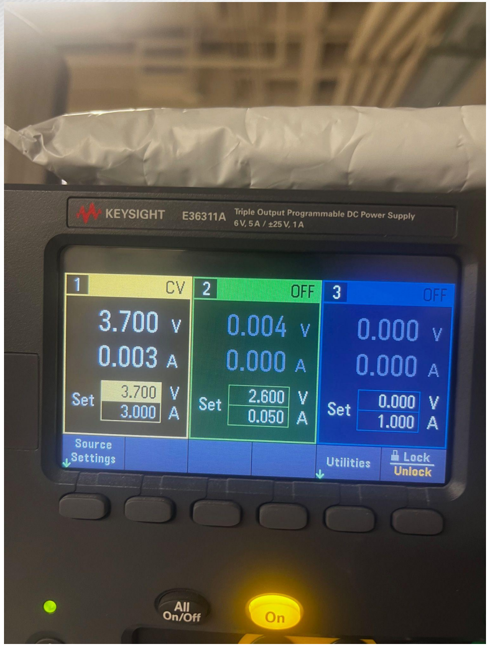


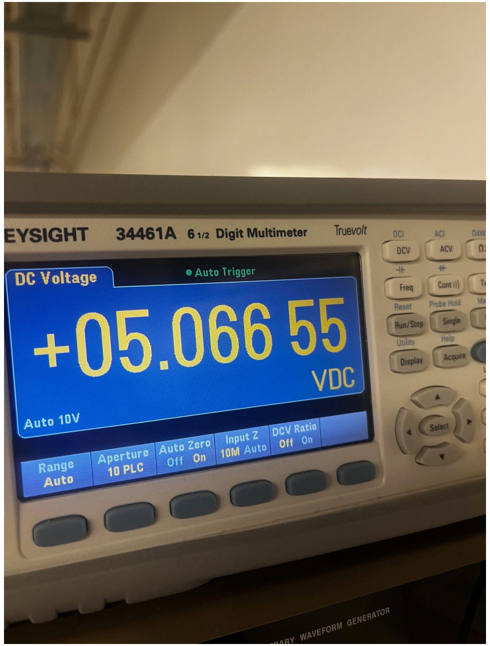
Image showing chg/dsg learning cycle

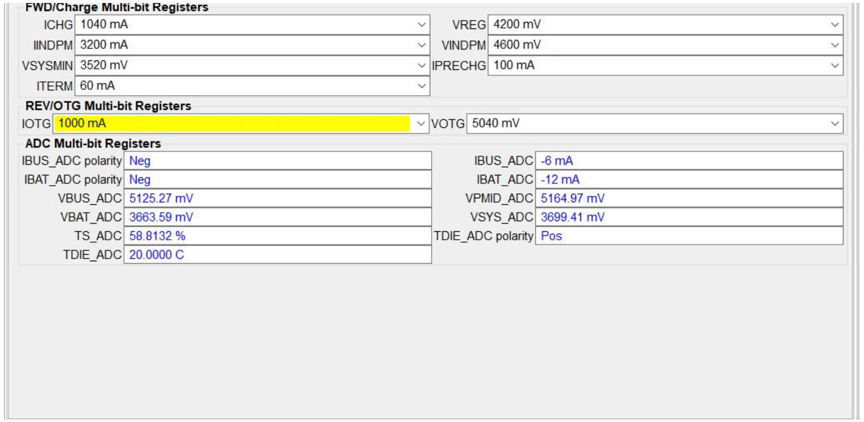
Cole Miller - Battery Charger

Accomplishments since last update 55 hrs of effort	Ongoing progress/problems and plans until the next presentation
Assembled and Validated High power path Removed MPPT from Solar Panel Validated I2C communication with EV2400 Validated USB-C IC	Fix issues with Battery Gauge Assemble USB-C and validate detection Validate control with MCU

Cole Miller - Battery Charger

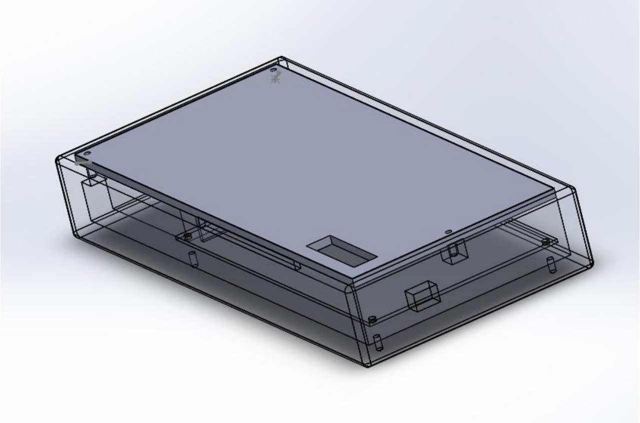


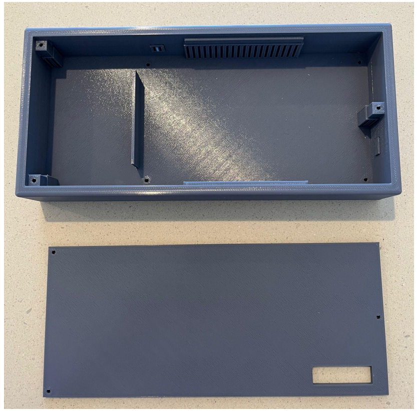




Camden Pomeroy - PMUX

Accomplishments since last update 60 hrs of effort	Ongoing progress/problems and plans until the next presentation
Created battery pack design using Solidworks 3D printed enclosure and verified fit Assembled power path on PCB Validated PMUX subsystem on PCB Validated high power path on fully integrated board	Assemble and validate MCU subsystem Finish validation testing





Jaein Cha - MCU

Accomplishments since last update 40 hrs of effort	Ongoing progress/problems and plans until the next presentation
- Debugged I2C - Tested 1:1 I2C functionality on the EVMs - Prototype of MPPT algorithm implemented - MPPT implementation tested with dummy signals for algorithm correctness - System control logic implemented and tested with dummy signals	- Test 1:N I2C functionality across all the EVMs we have - Test 1:N I2C functionality on board - Integrate MPPT with the I2C and test with the actual signals - Validate system controls on the real system

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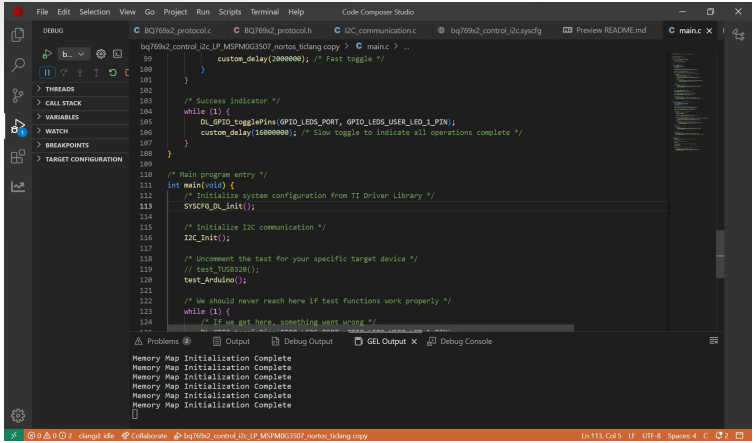
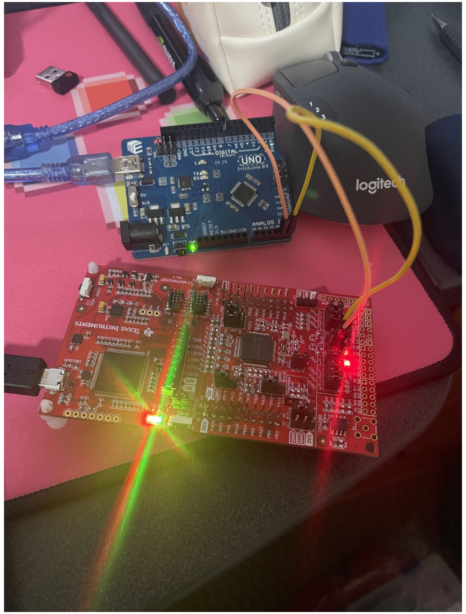
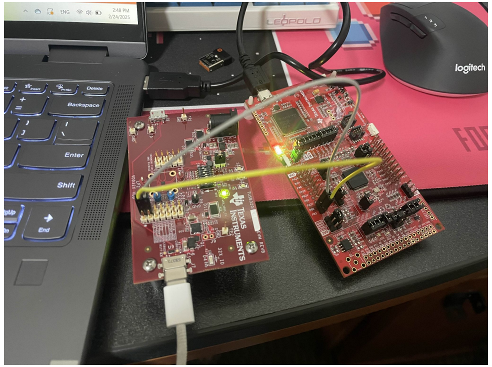
Jaein Cha - MCU







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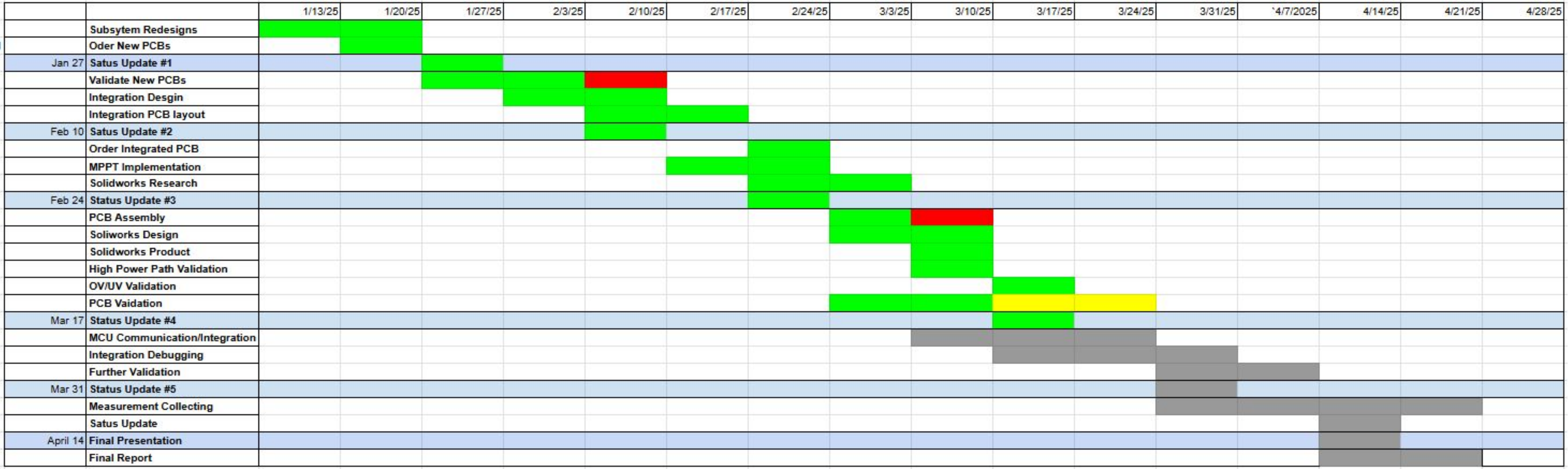
Execution Plan



System Validation Plan

Number	Validation Test	Input	Expected Output	Results	Status	Engineer	Reference Number
1	PP1 Operational	4-6V 2.5-3.5A on Vin (PP1)	4-6V 2.5-3.5A on Vout (PP1)	As expected		Camden Pomeroy	3.2.3.2.2
2	PP2 Operational	4-6V 2.5-3.5A in Vin (PP2)	4-6V 2.5-3.5A on Vout (PP2)	As expected		Camden Pomeroy	3.2.3.2.2
3	PP1 PG (Good signal)	If Vin > 2V	PG signal is read Low from MCU	Not tested		Camden Pomeroy	6.1.1
4	PP1 PG (Bad signal)	if Vin < 2V	PG signal is read High from MCU	Not tested		Camden Pomeroy	6.1.1
5	PP2 PG (Good signal)	If Vin > 2V	PG signal is read Low from MCU	Not tested		Camden Pomeroy	6.1.1
6	PP2 PG (Bad signal)	if Vin < 2V	PG signal is read High from MCU	Not tested		Camden Pomeroy	6.1.1
7	PP1 UVLO Operational	if Vin < UVLO	No output on VBUS	As expected		Camden Pomeroy	3.2.3.1
8	PP2 UVLO Operational	if Vin < UVLO	No output on VBUS	As expected		Camden Pomeroy	3.2.3.1
9	PP1 OVLO Operational	if Vin > OVLO	No output on VBUS	As expected		Camden Pomeroy	3.2.3.1
10	PP2 OVLO Operational	if Vin > OVLO	No output on VBUS	As expected		Camden Pomeroy	3.2.3.1
11	SEL USBC	SEL is sent low from MCU	PP1 is active and PP2 is not	Not tested		Camden Pomeroy	6.1.4
12	SEL Solar	SEL is sent high from MCU	PP2 is active and PP1 is not	Not tested		Camden Pomeroy	6.1.4
13	PGTH Threshold	Test Voltage of PGTH	PGTH 3.5V	As expected		Camden Pomeroy	3.2.3.2.2
14	SOC Display	SOC from 0-100%	1,2,3,4, or 5 LEDs on as expected	Not tested		Jaein Cha	3.2.3.2.1
15	I2C functionality (1:1, EVM)	Run I2C test code	It reads the default register value	As expected		Jaein Cha	6.1
16	I2C functionality (1:1, on-board)	Run I2C test code	It reads the default register value	Not tested		Jaein Cha	6.1
17	I2C functionality (1:N, EVM)	Run I2C test code	It reads the default register values	Not tested		Jaein Cha	6.1
18	I2C functionality (1:N, on-board)	Run I2C test code	It reads the default register values	Not tested		Jaein Cha	6.1
19	I2C - initialization	3.3V Power supply to the po	Setup the communication correctl	Not tested		Jaein Cha	6.1
20	I2C - DFP configuration	DFP plug is connected to th	Read TUSB I2C address and valu	Not tested		Jaein Cha	6.1
21	I2C - UFP configuration	UFP plug is connected to th	Read TUSB I2C address and valu	Not tested		Jaein Cha	6.1
22	I2C - DRP configuration	DRP plug is connected to th	Read TUSB I2C address and valu	Not tested		Jaein Cha	6.1
23	Reset	Reset button pressed	MCU resets and stop working whi	Not tested		Jaein Cha	3.2.3.2.1
24	MPPT (simulation)	Various voltages	maintain the power as close as 15	As expected		Jaein Cha	6.2
25	MPPT (on-board)	Various voltages	maintain the power as close as 15	Not tested		Jaein Cha	6.2
26	SOP Display	SOP button	Lights up all the LEDs when SOP	Not tested		Jaein Cha	3.2.3.2.1
27	Standard Operation	5V 3A on VBUS	around 5V 3A on VBUS	4.2V 3.2A on VBAT		Cole Miller	2.1
28	USB OTG Function	4.2V 3.2A on VBAT	around 4.2V 3.2A on VBAT	5V 1.5A on VBUS		Cole Miller	2.2
29	ILIM Register Functionality	If input current >1A	output current >1A	Output 1A		Cole Miller	2.3
30	VLIM Functionality	If Vin > 4.2V	Vo > 4.2V	Output 4.2V		Cole Miller	2.4
31	Solar Panel VOC	Solar panel Voc < 18V	Solar panel Voc < 18V	As expected		Cole Miller	6.3.1
32	Battery Charger I2C	Write to registers. Write cha	Read from said registers	Not tested		Cole Miller	6.1.2
33	Solar Panel Power	regular usage in sunny day	output 15W	As expected		Cole Miller	6.3.12
34	Battery Guage I2C	Write to registers. Read batt	Read from said registers	Not tested		Patrick Wang	6.1.3
35	Battery Lifespan	Battery connected to load	The lifespan is at least 3 hours	Not tested		Partick Wang	3.2.1.2
36	Power Consumption	Normal Operating Condition	Maximum power consumption is b	Not tested		Patrick Wang	3.2.1.2
37	Battery SOH	Discharge batteries from 10	Batteries stay in within an appropi	Not tested		Patrick Wang	5.1
38	High Power Path Charge	5V 3A from power MUX	Charges Battery Pack	As expected		Patrick, Cole, Camden	3.2.3.2.2
39	High Power Path Discharge	Load Connected to USBC	Battery Pack Discharges 15W	Not tested		Patrick , Cole, Camden	3.2.3.2.2
40	USBC Charging	USBC plug from wall outlet	Battery Pack Charges	Not tested		Patrick Wang	6.2.1

Final slide – Questions?